UNIVERSITY OF GREENWICH

Alliance with FPT Education

UNIVERSITY OF GREENWICH

COMP1682 – Final Year Projects

Proposal

Student name	TRAN TUAN AN
ID number (00xxxxxxx)	001555809
Lecturer/Tutor name	NGUYEN KIM KHANH
Student submission date	29/4/2026

Table of Contents

1. Overview 3
2. Aim 3
3. Related works / Literature review 3
4. Objectives 4
   - 4.1 Product Catalog Management 4
   - 4.2 Bulk Ordering & Payment Processing 4
   - 4.3 Delivery & Tracking System 4
   - 4.4 User & Client Management 4
   - 4.5 Project Framework or Any Methodology used. 5
5. Legal, Social, Ethical and Professional 6
   - 5.1. Legal 6
   - 5.2. Social 6
   - 5.3. Ethical 6
   - 5.4. Professional 6
6. Planning (see appendix A) 6
7. Initial References 8

References 8

Declaration of AI Use 9

COMP1682.1 Project Proposal
E-commerce Distribution Platform for Packaged Food and Beverages in Institutional Canteens

TRAN TUAN AN
BSc (Hons) Computing
001555809

1. Overview

In the rapidly expanding field of e-commerce, institutional food services such as school canteens, university cafeterias, and factory dining halls often face challenges in sourcing packaged food and beverages efficiently. Traditional distribution channels are fragmented, lack transparency, and rely heavily on manual processes, which can lead to delays, inconsistent supply, and limited product variety. This project proposes the development of a specialized e-commerce distribution platform for packaged food, drinks, and confectionery products, designed to serve institutional buyers rather than individual consumers. The platform will provide a centralized system where canteens and cafeterias can browse product catalogs, place bulk orders, track deliveries, and manage invoices seamlessly. By positioning the company as a distributor, the solution will streamline supply chains, reduce operational costs, and ensure reliable access to safe, high-quality packaged goods. The project aims to modernize the distribution process, enhance trust between suppliers and institutional buyers, and create a scalable model that can expand to additional categories of packaged products in the future (Material, 2026).

Keywords: e-commerce, food distribution, packaged goods, school canteen, university cafeteria, factory dining hall, bulk ordering.

2. Aim

The aim of this project is to develop a comprehensive e-commerce distribution platform for packaged food and beverages, targeting institutional buyers such as school canteens, university cafeterias, and factory dining halls. Upon completion, the system will provide a centralized solution that enables bulk ordering of confectionery, snacks, drinks, and other packaged products. It will support transparent inventory management, secure payment processing, and efficient delivery tracking. By positioning the company as a distributor rather than a direct retailer, the platform will streamline supply chains, reduce operational costs, and ensure reliable access to safe, high-quality products for large-scale buyers. This project will not only modernize the distribution process but also create a scalable model that can expand to additional categories of packaged goods, enhancing efficiency and trust across institutional food services (Material, 2026).

3. Related works / Literature review

Several e-commerce platforms currently dominate the market, such as Amazon, Shopee, Lazada, and Tiki, which provide a wide range of consumer products. While these platforms are effective for individual retail customers, they are not optimized for institutional buyers who require bulk ordering and specialized distribution services. Existing food distribution systems often rely on traditional offline processes, resulting in inefficiencies, limited transparency, and poor scalability.

Research into supply chain management systems highlights the importance of centralized platforms that integrate product catalogs, order tracking, and invoice management. Technologies such as secure

payment gateways, inventory management systems, and recommendation engines have been widely applied in retail e-commerce but remain underutilized in institutional food distribution.

Similar products include wholesale ordering systems used by supermarkets or B2B platforms like Alibaba, which focus on bulk transactions. However, these platforms are often too broad and lack customization for specific institutional contexts such as school canteens or factory cafeterias.

This project differentiates itself by focusing on packaged food and beverages distribution for institutional buyers, combining the efficiency of modern e-commerce with the reliability and transparency required in large-scale food services (Material, 2026).

4. Objectives

4.1 Product Catalog Management

Outcome: A structured system to manage packaged food, drinks, and confectionery products, including categories, stock levels, and pricing.

Activities:

4.1.1 Design and implement product database schema.

4.1.2 Develop admin interface for adding and updating products.

4.1.3 Integrate stock monitoring and alerts.

4.2 Bulk Ordering & Payment Processing

Outcome: A seamless ordering system that supports bulk purchases and secure payment methods suitable for institutional buyers.

Activities:

4.2.1 Implement bulk order functionality with quantity-based pricing.

4.2.2 Integrate secure payment gateways (e-wallets, bank transfer, COD).

4.2.3 Develop invoice generation and management module.

4.3 Delivery & Tracking System

Outcome: A transparent delivery tracking system that ensures timely distribution of products to schools, universities, and factories.

Activities:

4.3.1 Build order tracking dashboard for buyers.

4.3.2 Integrate logistics API for delivery status updates.

4.4 User & Client Management

Outcome: A role-based access system for administrators, distributors, and institutional buyers.

Activities:

4.4.1 Implement user authentication and role assignment.

4.4.2 Develop client profile management (canteens, cafeterias, factories).

4.4.3 Create reporting tools for sales and customer insight.

Effort Estimation (WBS)

To estimate the project effort, we applied a Work Breakdown Structure (WBS) to divide each objective into smaller, manageable tasks. Time allocations were then determined using a mix of expert judgment and analogy-based estimation, drawing on prior experience with similar e-commerce systems and comparing tasks to past projects of comparable scope.

Effort was expressed in weeks of work (40 hours per week), with smaller tasks calculated in fractions of a week. Estimates were aligned with the sequential phases of the Waterfall model, resulting in a total of about 18 weeks. A buffer was added to complex tasks to account for unforeseen challenges, ensuring the schedule remains realistic (Material, 2026).

Objective	Activities	Estimation	Total
Product Catalog Management	Design & implement product database schema	0.5 weeks	4 weeks
	Admin interface for product updates	2 weeks	
	Stock monitoring & alerts	1.5 weeks	
Bulk Ordering & Payment	Bulk order functionality	0.5 weeks	4 weeks
	Secure payment gateway integration	1.5 weeks	
	Invoice generation & management	2 weeks	
Delivery & Tracking System	Order tracking dashboard	1.5 weeks	3.5 weeks
	Logistics API integration	2 weeks	
User & Client Management	Authentication & role assignment	3 weeks	6.5 weeks
	Client profile management	1.5 weeks	
	Reporting tools	2 weeks	
Total hours			18 weeks

Table 1: Effort estimation of objective

4.5 Project Framework or Any Methodology used.

This project will adopt the Waterfall model as the primary development methodology. The Waterfall approach is suitable because the requirements of the e-commerce distribution platform are clearly defined, the scope is stable, and the project involves sequential phases that can be planned in detail. Each stage must be completed before moving to the next, ensuring structured progress and minimizing risks of misalignment (Material, 2026).

The development process will include the following phases:

- **Requirement Analysis (2 weeks):** Collect and document detailed requirements from institutional buyers such as school canteens, university cafeterias, and factory dining halls.
- **System Design (2 weeks):** Create system architecture, database schema, and user interface designs based on the requirements.
- **Implementation (8 weeks):** Develop core modules including product catalog management, bulk ordering, payment integration, delivery tracking, and reporting dashboard.
- **Testing (2 weeks):** Conduct unit testing, integration testing, and system testing to ensure reliability and performance.
- **Deployment (1 week):** Launch the platform for use by selected institutional buyers.
- **Maintenance (2 weeks):** Provide bug fixes, updates, and ensure scalability for future product categories.

The Waterfall model ensures a clear timeline, structured documentation, and predictable outcomes, making it appropriate for a project with well-defined objectives and institutional stakeholders (Material, 2026).

5. Legal, Social, Ethical and Professional

5.1. Legal

The project must comply with relevant e-commerce and consumer protection laws, including regulations on online transactions, invoicing, and data privacy. Compliance with frameworks such as **GDPR** or local equivalents is essential to safeguard customer information. Additionally, food distribution must adhere to health and safety standards, ensuring that packaged products meet quality certifications and labeling requirements (Material, 2026).

5.2. Social

The platform will contribute to improving institutional food services by providing reliable access to safe and diverse packaged goods. By streamlining distribution, schools, universities, and factories can offer better food options to students and workers. This enhances convenience, reduces supply chain inefficiencies, and supports healthier consumption habits (Material, 2026).

5.3. Ethical

Transparency in pricing, product quality, and delivery processes is critical. The system must prevent unfair practices such as hidden fees or misleading product descriptions. Ethical responsibility also includes ensuring that all distributed products are safe, properly labeled, and sourced from reputable suppliers (Material, 2026).

5.4. Professional

The project will be developed following professional software engineering standards, including structured documentation, version control, and rigorous testing. Professional conduct requires maintaining clear communication with stakeholders, ensuring accountability in system design, and providing long-term support and maintenance. The platform will be designed to scale responsibly, with a focus on sustainability and trustworthiness in institutional food distribution (Material, 2026).

6. Planning (see appendix A)

The project will follow the Waterfall model, which divides the development process into sequential phases. Each phase must be completed before moving to the next, ensuring clarity, structured documentation, and predictable outcomes (Material, 2026).

Development Phases and Timeline:

Requirement Analysis (Weeks 1–2):

- Gather requirements from institutional buyers (school canteens, university cafeterias, factory dining halls).
- Document functional and non-functional requirements.
- Define scope and constraints of the system.

System Design (Weeks 3–4):

- Create system architecture diagrams.
- Design database schema for product catalog, orders, and clients.
- Develop user interface prototypes for buyers and administrators.

Implementation (Weeks 5–12):

- Develop product catalog management module.
- Implement bulk ordering and secure payment system.
- Build delivery tracking and logistics integration.
- Create client management and reporting dashboard.

Testing (Weeks 13–14):

- Conduct unit testing for each module.
- Perform integration testing across modules.
- Carry out user acceptance testing with selected institutional buyers.

Deployment (Week 15):

- Launch the platform for pilot use.
- Train administrators and provide documentation.

Maintenance (Weeks 16–17):

- Fix bugs and optimize performance.
- Provide updates and scalability improvements.

Project Management Tools:

- The schedule will be visualized using MS Project with a Gantt chart to track tasks dependencies, and milestones.
- Each phase will have clear deliverables and deadlines to ensure accountability.

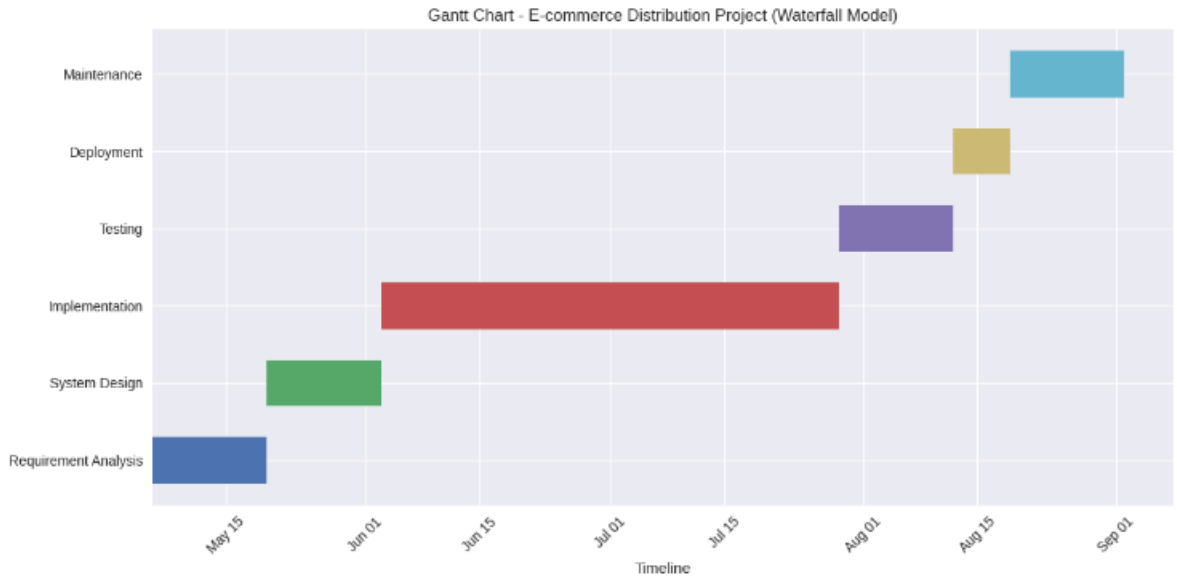


Figure1: Gantt Chart

7. Initial References

References

Material, G., 2026. [Online]
Available at: https://librarysearch.gre.ac.uk/client/en_GB/gre

Declaration of AI Use

Please append this page to the end of your assignment when you have used AI during the process of undertaking the assignment to acknowledge the ways in which you have used it.

I have used AI while undertaking my assignment in the following ways:

- To develop research questions on the topic: YES
- To create an outline of the topic: NO
- To explain concepts: NO
- To support my use of language: YES
- To summarise the following articles/resources: YES